Next Day Wildfire Spread Prediction

Phase 1: per-pixel boosted trees with engineered spatial features

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Problem: which pixels burn tomorrow?

- Input: 12 channels on day t for a 64×64 km patch at 1 km (weather, terrain, vegetation, population, today's fire)
- Output: probability that each pixel burns on day $t{+}1$
- Fire is rare: 1.25% of test pixels (1 in 80); cloud/smoke pixels (-1) are ignored
- Accuracy fails: "always no fire" scores 98.7%, our model 97.3%. So we use **PR AUC**: precision vs recall on the fire class over all thresholds (random model = 0.0125, the fire rate)

Model	Accuracy	Fire found
Always "no fire"	98.7%	0%
Our tuned XGBoost	97.3%	53%

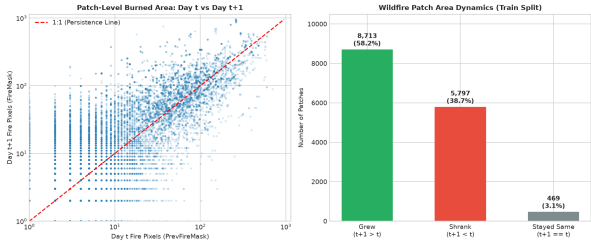
Related work at a glance

- Huot et al. 2022 built our dataset: pixel models up to 0.225 PR AUC, whole-patch CNN 0.284
- Zhou et al. 2025: XGBoost on raw pixels only 0.06 PR AUC. Features, not the algorithm, decide what trees can do
- Other papers use different crops, labels or metrics, so their scores are context only

Paper	Method	Data	Metric	Result
Hodges & Lattimer 2019	CNN (DCIGN)	simulated fires	F vs. simulation	0.93
Radke et al. 2019	CNN + weather	Rocky Mts., 30 m	precision / recall	3.6% / 91.1%
Gerard et al. 2023	U-Net / UTAE	VIIRS, 375 m	AP	0.341 / 0.372
Huot et al. 2022	persistence	NDWS, 1 km	PR AUC	0.115
Huot et al. 2022	LR + 8 neighbours	NDWS	PR AUC	0.198
Huot et al. 2022	RF + 8 neighbours	NDWS	PR AUC	0.225
Huot et al. 2022	CNN autoencoder	NDWS	PR AUC	0.284

The dataset

- 18 545 patches of US fires (2012–2020); split by calendar week 8:1:1 → 14 979 / 1 877 / 1 689
- Labels: 1 fire, 0 no fire, −1 unknown (2.3–2.8% of labels) → dropped from training and metrics
- Fires change overnight: 58% grow, 39% shrink (train). Persistence misses growth and over-predicts shrinkage



Burned area per train patch on day t vs $t{+}1$; dashed line = persistence.

Data traps: artefacts and a south-up grid

Export artefacts

- Wind direction from $-505\,870^\circ$ to $37\,736^\circ$ $\rightarrow$ wrapped (θ mod 360)
- Temperatures from -445 K to $1\,230$ K $\rightarrow$ clipped to $[253, 330]$ K
- Negative rain in 9.8% of pixels $\rightarrow$ clipped to ≥ 0
- Bounds are physical constants, not fitted on data $\rightarrow$ no leakage

Patches are stored south-up (not documented)

- North-facing slopes are greener in the Northern Hemisphere: slopes facing the last row have NDVI 0.596 vs 0.555 $\rightarrow$ the last row is north
- Downwind/upwind next-day fire ratio at $\geq$ 8 m/s: 3.1 with south-up, 0.37 with north-up (fire spreading into the wind is impossible)
- Every wind feature depends on it: one constant, `ROW_TO_NORTH = 1`

Assumed orientation	Strong-wind ratio
North-up	0.37
South-up	3.1

Found because our first downwind check gave a physically impossible answer.

Method: from TFRecords to a per-pixel model

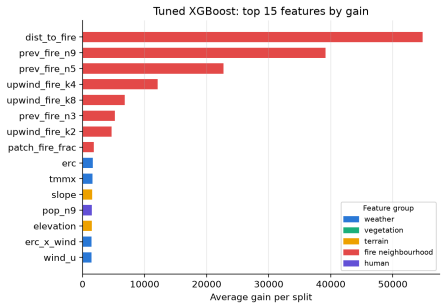


- TFRecords parsed without TensorFlow into $(N, 12, 64, 64)$ arrays
- Artefacts clipped to fixed physical bounds; wind direction wrapped (θ mod 360)
- South-up orientation used by every direction feature (slide 5)

- Training table: every fire pixel + 5:1 no-fire sample = 3.95 M rows; `scale_pos_weight` = 5
- 20 random configurations per model, early stopping on eval, best by eval PR AUC
- Decision threshold: best F1 on eval, applied unchanged to test

Features: 12 raw + 23 engineered, in five groups

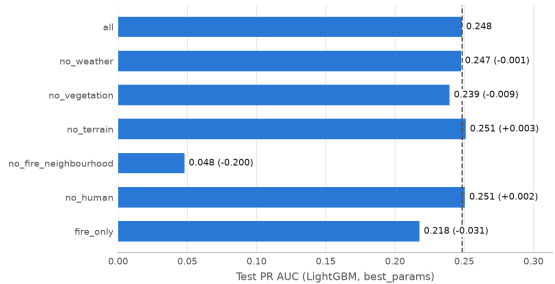
Group	#	Engineered examples
Fire neighbourhood	10	fire density (3/5/9 px), distance, fire upwind
Weather	14	wind u/v, VPD, min. rel. humidity, ERC $\times$ wind
Vegetation	4	local NDVI mean, std, anomaly
Terrain	5	slope, aspect, relative elevation
Human	2	population, 9×9 mean



- `dist_to_fire`: univariate AUC 0.866 vs 0.593 for the raw fire mask
- Most XGBoost gain is in fire features: `dist_to_fire` 32%, `prev_fire_n9` 23%, upwind 13.5%

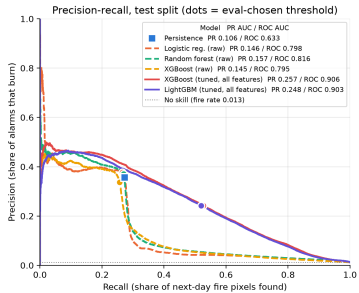
From EDA to features, and what each group is worth

- 40% of new fire is more than 5 km from day-t fire $\rightarrow$ added `upwind_fire_k8` (ranks 5th of 35 by XGBoost gain)
- Ablation: dropping fire-neighbourhood features drops PR AUC 0.248 $\rightarrow$ 0.048; vegetation is the most useful environmental group (-0.009)
- Fire features alone: 0.218; environment adds $+0.031$ on top



Results: same algorithm, nearly double the score

Model	PR AUC	ROC AUC	Recall	F1
Persistence	0.106	0.633	0.273	0.309
Logistic regression (raw)	0.146	0.798	0.266	0.312
Random forest (raw)	0.157	0.816	0.270	0.312
XGBoost (raw)	0.145	0.795	0.257	0.292
XGBoost (tuned, 35 feat.)	**0.257**	**0.906**	**0.526**	**0.332**
LightGBM (tuned, 35 feat.)	0.248	0.903	0.520	0.331

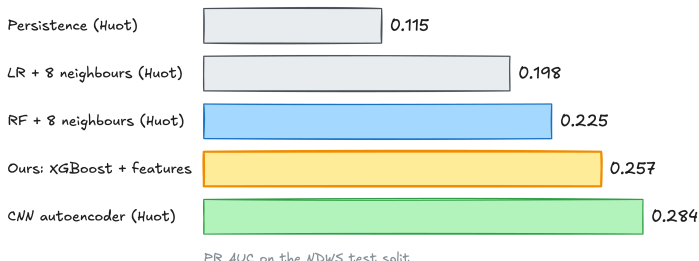


- XGBoost raw 0.145 → with our features 0.257: the gain comes from the features
- Recall doubles (26–27% → 53%) at 24% precision; 39% of fire found at 1% FPR, 68% at 5% FPR

Test split; recall and F1 at the threshold chosen on eval.

Where we stand

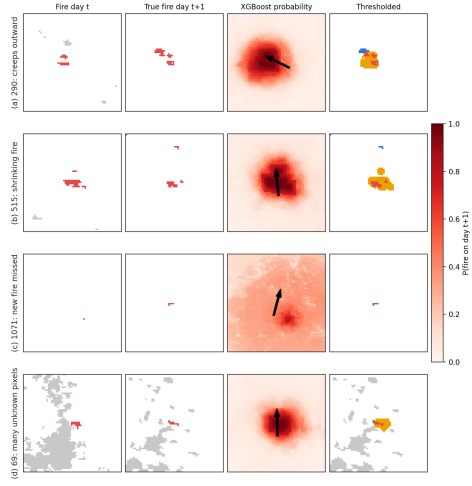
- 0.257 PR AUC; 95% interval over fire events 0.19–0.33
- Huot's RF (0.225) and CNN (0.284) both lie inside it → we sit between them, with no claim of being better
- Raw XGBoost elsewhere: 0.06 (Zhou 2025); recent U-Nets about 0.35
- Tuning moves PR AUC by about 0.01, spatial features by about 0.20



What the model learned

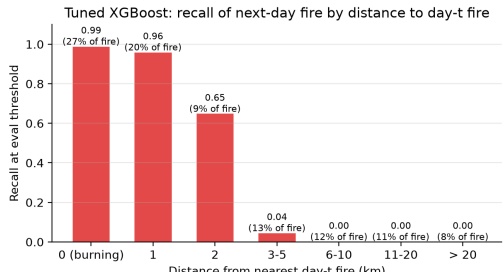
- Predicts a halo around today's fire, leaning downwind
- SHAP: fire-neighbourhood features dominate (2.29 log-odds vs 0.40 weather, 0.22 vegetation)
- Physically sensible signs: near fire, dry air (VPD), strong wind, fire upwind → higher risk. ERC is the one exception: it acts as a proxy for region/season
- Pushes risk downwind; train data agrees: the downwind/upwind ratio grows from 1.135 (calm) to 3.100 (≥ 8 m/s)

(a) creeps outward: caught; (b) shrinking fire: false-alarm ring; (c) new fire away from the front: missed; (d) many unknown pixels. Red = hit, orange = false alarm, blue = missed.



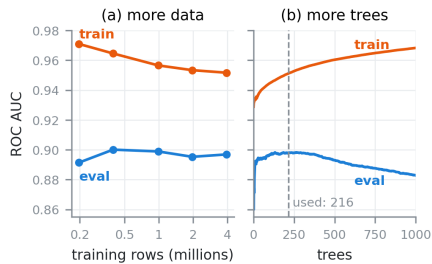
Where it fails

- False alarms around dying fires: 88.8% within 2 km of today's fire; only 35.3% of burning pixels still burn tomorrow
- Misses long-range spread: recall 0.988 at the fire, 0.648 at 2 km, 0 beyond 5 km (64% of missed fire)
- Per fire: 1 689 test patches = only 257 fires. The ranking holds per fire (mean PR AUC 0.187 vs 0.074–0.101), but the eval threshold finds fire in only 48% of fires vs 61% for persistence
- Fixed-radius halo, no fire-front shape → Phase 2



Can we trust 0.257?

- Reproduced from raw TFRecords on Kaggle: every score within 0.002
- Bootstrap over 257 fires: 0.19–0.33. Paired: XGB > LGBM in 99% of resamples, > raw RF in 100%
- Calibration: one closed-form correction (odds ÷ 90) gives a mean of 0.0107 vs 0.0125 observed; threshold 0.915 ≈ true p of 0.107
- Learning curve: 5% of the training data → 0.204, 100% → 0.207. Limited by bias, not variance → more data or tuning won't help
- Location overlap: 30% of test patches share a location with a train patch; on new locations only the ranking is unchanged (XGB 0.243 > LGBM 0.234 > RF 0.145 > persistence 0.097)



Conclusion

- 23 engineered spatial features take XGBoost from 0.145 to **0.257** PR AUC (persistence 0.106), between Huot's RF and CNN
- The signal is spatial: removing fire-neighbourhood features drops PR AUC to 0.048; tuning moves it by about 0.01
- Robust: reproduced, holds per fire and on unseen locations, physically sensible (SHAP, downwind)
- Limited by representation: a fixed-radius halo misses spread beyond 5 km and over-predicts dying fires

Side finding: the dataset is stored south-up, which every wind feature depends on.

Future work (Phase 2)

- **Whole-patch models:** U-Net / CNN that see the shape of the fire front instead of a fixed-radius halo
- **Imbalance-aware losses:** focal loss or class-weighted segmentation loss
- **Multi-day inputs:** fire history over several days (WildfireSpreadTS)
- **Location-disjoint splits:** 30% of test patches share their location with a training patch; a split that keeps locations apart would give a stricter test
- **Uncertain labels:** better handling of cloud and smoke (-1) pixels than simply ignoring them

Thank you

Questions?

Code, report and figures: `github.com/harshilv17/next-day-wildfire-spread`